

Sustainable Quantum Federated Learning for Medical Image Analysis: A Four-Axis Accounting Protocol

Problem

QFL paper A	params: 8k accuracy: 89%
QFL paper B	comm: 12 MB F ₁ : 0.81
QFL paper C	qubits: 6 layers: 3 shots: 1024

Sustainability claims reported in incomparable units.

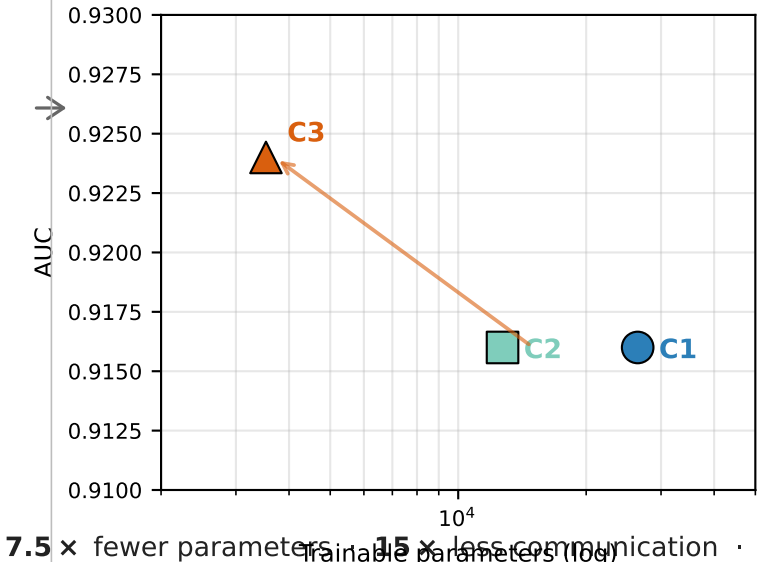
SUSQA Protocol

Four axes, each tied to a real sustainability cost

P Parameters trainable count	C_{total} Communication MB per FL run
Q Quantum cost gate · shots · samples	$\eta(p)$ Noise robustness AUC retention

Result

PneumoniaMNIST under FL IID



7.5 × fewer parameters · 15 × less communication · higher AUC